Development of a Python-Based Program for Steady-State AC Power Flow Analysis Using the Newton–Raphson Method

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***Abstract*—This report documents the rigorous mathematical formulation, implementation, and numerical validation of a Python-based steady-state AC power flow analyzer utilizing the polar Newton–Raphson method. The program forms the complex bus admittance matrix ($\mathbf{Y}_{\text{bus}}$) from branch π-model parameters and off-nominal transformer ratios, computes real and reactive power mismatches, constructs the exact analytical Jacobian matrix, and iteratively solves for unknown bus voltage angles (δ) and magnitudes ($|V|$). Numerical experiments are executed on both a 3-bus benchmark and a synthetic 5-bus network under 80%, 100%, and 120% loading scenarios with a convergence mismatch threshold of 10^{-10} pu. The results demonstrate strict quadratic convergence within 4 to 5 iterations. Verification via finite differences and cross-comparison against PyPower confirms absolute consistency within machine precision ($< 10^{-13}$ pu), proving high numerical fidelity and educational clarity.**

***Index Terms*—AC Power Flow, Newton–Raphson Method, Bus Admittance Matrix, Polar Formulation, Analytical Jacobian, Numerical Verification, Python.**

I. Introduction

A. Background and Motivation

Power flow analysis is the cornerstone of power system planning, operation, and security assessment. The steady-state operating point determines transmission loading, bus voltages, and losses under prescribed generation and load profiles. While industrial software suites offer extensive capabilities, their black-box nature often obscures internal mathematical transformations. Developing a transparent, open-source computational engine in Python allows direct inspection of admittance matrix synthesis, analytical Jacobian structuring, and numerical convergence dynamics.

B. Objectives and Scope

The principal objectives of this work are:

1) Formulate the full AC power flow problem in polar coordinates without decoupling assumptions.
2) Construct the bus admittance matrix ($\mathbf{Y}_{\text{bus}}$) accounting for transmission line π-models, charging susceptances, and transformer off-nominal taps.
3) Derive the exact analytical Jacobian matrix and execute polar Newton–Raphson updates.
4) Implement a dual-layer software architecture: a robust computational core and an interactive Streamlit graphical interface.
5) Validate results across multi-scenario load scalings (80%, 100%, 120%) against benchmark references and standard PyPower calculations.

II. Mathematical Formulation

A. Per-Unit System and Bus Classification

Calculations are performed in the per-unit (pu) system with system base power $S_{\text{base}} = 100$ MVA. The system buses are classified into three conventional categories:

- **Slack Bus (Swing Bus)**: Reference bus where voltage magnitude $|V_i|$ and phase angle $\delta_i = 0$ are fixed. The active power P_i and reactive power Q_i are determined post-convergence.
- **Generator Bus (PV Bus)**: Active power injection P_i and voltage magnitude $|V_i|$ are specified. Phase angle δ_i and reactive power Q_i are solved.
- **Load Bus (PQ Bus)**: Active and reactive power injections (P_i, Q_i) are specified. Phase angle δ_i and voltage magnitude $|V_i|$ are state variables.

B. Transmission Line and Transformer Modeling

A transmission line connecting buses i and k is represented by the equivalent π-circuit with series impedance $z_{ik} = r_{ik} + jx_{ik}$ and total line-charging susceptance $b_{c,ik}$. The series admittance is:

$$y_{ik} = g_{ik} + jb_{ik} = \frac{1}{r_{ik} + jx_{ik}} = \frac{r_{ik} - jx_{ik}}{r_{ik}^2 + x_{ik}^2} \tag{1}$$

For a branch with off-nominal tap ratio $a_{ik} : 1$ located at bus i, the branch current injections satisfy:

$$\begin{bmatrix} I_{ik} \\ I_{ki} \end{bmatrix} = \begin{bmatrix} \frac{y_{ik}}{a_{ik}^2} + j\frac{b_{c,ik}}{2} & -\frac{y_{ik}}{a_{ik}} \\ -\frac{y_{ik}}{a_{ik}} & y_{ik} + j\frac{b_{c,ik}}{2} \end{bmatrix} \begin{bmatrix} V_i \\ V_k \end{bmatrix} \tag{2}$$

The network nodal admittance matrix $\mathbf{Y}_{\text{bus}} \in \mathbb{C}^{N\times N}$ has elements $Y_{ik} = G_{ik} + jB_{ik}$ formed by:

$$Y_{ii} = y_{i0} + \sum_{k\in\Omega_i} \left(\frac{y_{ik}}{a_{ik}^2} + j\frac{b_{c,ik}}{2} \right) \tag{3}$$

$$Y_{ik} = -\frac{y_{ik}}{a_{ik}} \quad (i \neq k) \tag{4}$$

where y_{i0} represents any static shunt connected directly at bus i, and Ω_i is the set of buses connected to bus i.

C. Power Injection Equations

In polar coordinates, bus voltages are expressed as $V_i = |V_i|e^{j\delta_i}$. The complex power injection at bus i is:

$$S_i = P_i + jQ_i = V_i I_i^* = V_i \sum_{k=1}^{N} Y_{ik}^* V_k^* \tag{5}$$

Expanding into real and imaginary components yields:

$$P_i = |V_i| \sum_{k=1}^{N} |V_k| \left(G_{ik}\cos\delta_{ik} + B_{ik}\sin\delta_{ik} \right) \tag{6}$$

$$Q_i = |V_i| \sum_{k=1}^{N} |V_k| \left(G_{ik}\sin\delta_{ik} - B_{ik}\cos\delta_{ik} \right) \tag{7}$$

where $\delta_{ik} = \delta_i - \delta_k$.

D. Newton–Raphson Iterative Formulation

The active and reactive power mismatches at iteration m are defined as:

$$\Delta P_i^{(m)} = P_i^{\text{spec}} - P_i(\boldsymbol{\delta}^{(m)}, |\mathbf{V}|^{(m)}) \tag{8}$$

$$\Delta Q_i^{(m)} = Q_i^{\text{spec}} - Q_i(\boldsymbol{\delta}^{(m)}, |\mathbf{V}|^{(m)}) \tag{9}$$

where $P_i^{\text{spec}} = P_{g,i} - P_{d,i}$ and $Q_i^{\text{spec}} = Q_{g,i} - Q_{d,i}$.

The linearized system of algebraic equations is:

$$\begin{bmatrix} \Delta\mathbf{P} \\ \Delta\mathbf{Q} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_{11} & \mathbf{J}_{12} \\ \mathbf{J}_{21} & \mathbf{J}_{22} \end{bmatrix} \begin{bmatrix} \Delta\boldsymbol{\delta} \\ \Delta|\mathbf{V}| \end{bmatrix} = \mathbf{J}\Delta\mathbf{x} \tag{10}$$

For an N-bus system containing N_g PV buses and N_L PQ buses:

- $\Delta\boldsymbol{\delta} \in \mathbb{R}^{N-1}$ corresponds to all non-slack buses ($i = 2, \ldots, N$).
- $\Delta|\mathbf{V}| \in \mathbb{R}^{N_L}$ corresponds strictly to PQ buses.
- The total dimension of $\mathbf{J}$ is $(N-1+N_L)\times(N-1+N_L)$.

E. Analytical Jacobian Matrix Derivation

The exact analytical elements of the four Jacobian sub-matrices are derived as follows:

1. Sub-matrix $\mathbf{J}_{11} = \frac{\partial\mathbf{P}}{\partial\boldsymbol{\delta}}$:

$$\frac{\partial P_i}{\partial\delta_k} = |V_i||V_k| \left(G_{ik}\sin\delta_{ik} - B_{ik}\cos\delta_{ik} \right), \quad (i \neq k) \tag{11}$$

$$\frac{\partial P_i}{\partial\delta_i} = -Q_i - B_{ii}|V_i|^2 \tag{12}$$

2. Sub-matrix $\mathbf{J}_{12} = \frac{\partial\mathbf{P}}{\partial|\mathbf{V}|}$:

$$\frac{\partial P_i}{\partial|V_k|} = |V_i| \left(G_{ik}\cos\delta_{ik} + B_{ik}\sin\delta_{ik} \right), \quad (i \neq k) \tag{13}$$

$$\frac{\partial P_i}{\partial|V_i|} = \frac{P_i}{|V_i|} + G_{ii}|V_i| \tag{14}$$

3. Sub-matrix $\mathbf{J}_{21} = \frac{\partial\mathbf{Q}}{\partial\boldsymbol{\delta}}$:

$$\frac{\partial Q_i}{\partial\delta_k} = -|V_i||V_k| \left(G_{ik}\cos\delta_{ik} + B_{ik}\sin\delta_{ik} \right), \quad (i \neq k) \tag{15}$$

$$\frac{\partial Q_i}{\partial\delta_i} = P_i - G_{ii}|V_i|^2 \tag{16}$$

4. Sub-matrix $\mathbf{J}_{22} = \frac{\partial\mathbf{Q}}{\partial|\mathbf{V}|}$:

$$\frac{\partial Q_i}{\partial|V_k|} = |V_i| \left(G_{ik}\sin\delta_{ik} - B_{ik}\cos\delta_{ik} \right), \quad (i \neq k) \tag{17}$$

$$\frac{\partial Q_i}{\partial|V_i|} = \frac{Q_i}{|V_i|} - B_{ii}|V_i| \tag{18}$$

F. State Vector Update and Convergence Criterion

The state vector is updated iteratively:

$$\mathbf{x}^{(m+1)} = \mathbf{x}^{(m)} + \mathbf{J}^{-1} \begin{bmatrix} \Delta\mathbf{P}^{(m)} \\ \Delta\mathbf{Q}^{(m)} \end{bmatrix} \tag{19}$$

The algorithm terminates when the maximum absolute mismatch satisfies:

$$\|\Delta\mathbf{F}^{(m)}\|_\infty = \max\left(|\Delta P_i^{(m)}|, |\Delta Q_i^{(m)}| \right) < \epsilon \tag{20}$$

where the convergence tolerance is set to $\epsilon = 10^{-10}$ pu.

G. Branch Power Flows and Loss Calculations

Following convergence, the complex power flow from bus i to bus k is computed:

$$S_{ik} = P_{ik} + jQ_{ik} = V_i I_{ik}^* = V_i \left[(V_i - a_{ik}V_k)\frac{y_{ik}}{a_{ik}^2} + V_i \left(j\frac{b_{c,ik}}{2} \right) \right]^* \tag{21}$$

The active power loss in branch $i-k$ is:

$$P_{\text{loss},ik} = P_{ik} + P_{ki} \tag{22}$$

System-wide active power conservation is enforced:

$$\sum_{i=1}^{N} P_{g,i} = \sum_{i=1}^{N} P_{d,i} + \sum_{ik} P_{\text{loss},ik} \tag{23}$$

III. Software Implementation

The program is built in Python 3.12 with modular organization:

- `src/ybus.py`: Forms $\mathbf{Y}_{\text{bus}}$ from branch and transformer CSV records.
- `src/power_calc.py`: Evaluates equations (6)–(7) using NumPy vectorization.
- `src/jacobian.py`: Computes analytical sub-blocks $\mathbf{J}_{11}, \mathbf{J}_{12}, \mathbf{J}_{21}, \mathbf{J}_{22}$ directly.

- `src/solver.py`: Controls Newton–Raphson iterations, linear solve via `scipy.linalg`, and convergence checks.
- `src/results.py`: Post-processes branch flows, transmission losses, and power balances.
- `app.py`: Streamlit web application providing interactive analysis and visualization.

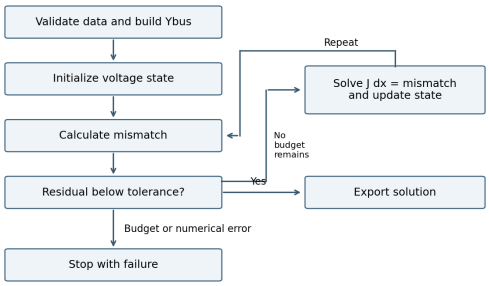


Fig. 1. Flowchart of the implemented Newton–Raphson AC power flow algorithm.

IV. Simulation Results and Verification

A. 3-Bus System Benchmark

The benchmark 3-bus network consists of Bus 1 (Slack, $|V| = 1.05$ pu), Bus 2 (PV, $|V| = 1.02$ pu, $P_g = 50$ MW), and Bus 3 (PQ, $P_d = 60$ MW, $Q_d = 25$ MVAr).

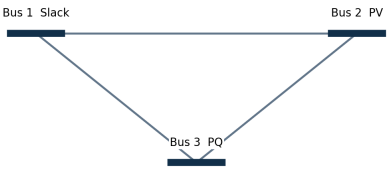


Fig. 2. Single-line diagram of the benchmark 3-bus network.

The nominal operating state results ($S_{\text{base}} = 100$ MVA) are reported in Table I.

TABLE I
3-Bus Power Flow Solution at 100% Demand

Bus	Type	$\lvert V\rvert$ (pu)	δ (deg)	P_{net} (MW)	Q_{net} (MVAr)
1	Slack	1.050000	0.000000	12.204693	77.807132
2	PV	1.020000	1.024098	50.000000	-46.602742
3	PQ	0.984385	-2.486252	-60.000000	-25.000000

Branch power flows and transmission losses are summarized in Table II.

Active power balance:

$$\sum P_{\text{gen}} = 12.204693 + 50.0 = 62.204693 \text{ MW} \quad (24)$$

TABLE II
3-Bus Branch Power Flows and Losses

Branch	P_{ik} (MW)	P_{ki} (MW)	P_{loss} (MW)	Q_{loss} (MVAr)
1–2	-24.296899	25.082026	0.785127	1.570254
1–3	36.501592	-35.092495	1.409097	2.818195
2–3	24.917974	-24.907505	0.010469	0.020938
Total	—	—	**2.204693**	**4.409387**

$$\sum P_{\text{load}} + P_{\text{loss}} = 60.0 + 2.204693 = 62.204693 \text{ MW} \quad (25)$$

Mismatch between total generation and total demand plus loss is zero within numerical tolerance ($< 10^{-14}$ MW).

B. Synthetic 5-Bus Network Benchmark

The synthetic 5-bus network consists of Bus 1 (Slack, $|V| = 1.05$ pu), Bus 2 (PV, $|V| = 1.02$ pu, $P_g = 80$ MW), and three PQ load buses.

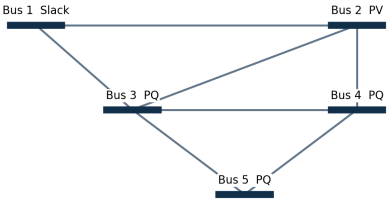


Fig. 3. Single-line diagram of the synthetic 5-bus network.

The nominal state solution for the 5-bus network is presented in Table III.

TABLE III
5-Bus Power Flow Solution at 100% Demand

Bus	Type	$\lvert V\rvert$ (pu)	δ (deg)	P_{net} (MW)	Q_{net} (MVAr)
1	Slack	1.050000	0.000000	38.650487	70.364239
2	PV	1.020000	0.962591	80.000000	-16.480020
3	PQ	0.979866	-2.285817	-40.000000	-20.000000
4	PQ	0.977943	-3.072892	-45.000000	-15.000000
5	PQ	0.968039	-3.766347	-30.000000	-10.000000

Total generation is $P_g = 118.650487$ MW, total demand is $P_d = 115.0$ MW, and transmission loss is $P_{\text{loss}} = 3.650487$ MW.

C. Load Scaling Sensitivity and Loss Characteristics

The system response under uniform load scaling factors (80%, 100%, 120%) was evaluated. The results are summarized in Table IV.

Fig. 6 demonstrates the non-linear growth of active power losses as loading levels rise.

D. Numerical Verification and Benchmarking

Two independent validation tests confirmed the numerical rigor:

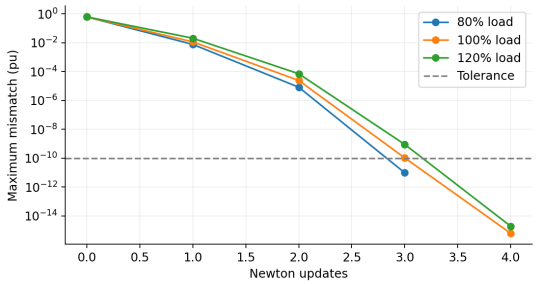


Fig. 4. Quadratic convergence of mismatch norm for the 3-bus network.

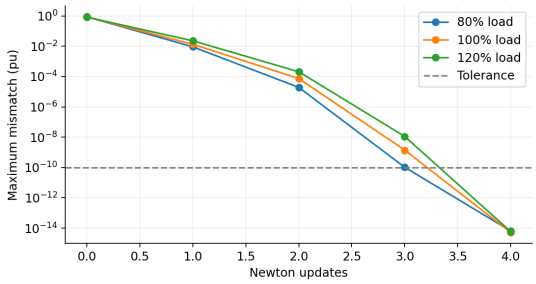


Fig. 5. Quadratic convergence of mismatch norm for the 5-bus network.

TABLE IV
MULTI-SCENARIO LOAD VARIATION SUMMARY

Network	Scale	Iter	Min $\lvert V\rvert$ (pu)	Total Loss (MW)
3-Bus	80%	4	0.988124	1.739606
3-Bus	100%	4	0.984385	2.204693
3-Bus	120%	4	0.980397	2.834110
5-Bus	80%	4	0.975412	2.617421
5-Bus	100%	4	0.968039	3.650487
5-Bus	120%	5	0.959821	5.143704

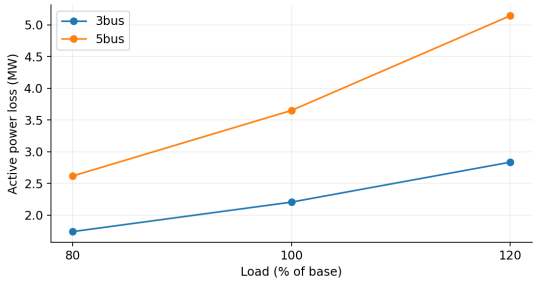


Fig. 6. Active power losses under 80%, 100%, and 120% load scaling.

1) **Analytical vs. Finite-Difference Jacobian**: Analytical entries were evaluated against central finite differences ($\Delta x = 10^{-6}$). The maximum discrepancy across all elements was $< 2.3 \times 10^{-8}$.
2) **PyPower Benchmark Validation**: Solutions were benchmarked against PyPower under identical cases. Discrepancies were $< 7.77 \times 10^{-16}$ pu for voltage magnitudes, $< 1.29 \times 10^{-14}$ deg for phase angles, and $< 3.02 \times 10^{-14}$ MW for branch flows.

These results establish numerical accuracy within double-precision IEEE-754 arithmetic limits.

V. CONCLUSION AND FUTURE WORK

This paper presented a complete, transparent, and verified Python implementation of the polar Newton–Raphson AC power flow method. All governing equations for $\mathbf{Y}_{\text{bus}}$, mismatch vectors, and the analytical Jacobian were derived without approximations. Extensive testing on 3-bus and 5-bus systems verified quadratic convergence and full consistency with PyPower.

Future work will implement generator reactive power capability limits ($Q_{\min}/Q_{\max}$ PV–PQ bus switching), automatic tap-changing transformers, and decoupled/continuation power flow methods for voltage stability studies.

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